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Module having a changing filler content rate in sealing resin layer

Abstract

A module includes a substrate, a first component, and a first sealing resin layer. The substrate includes a first principal surface. The first component is mounted on the first principal surface. The first sealing resin layer contains a filler containing an inorganic oxide as a main component. The first sealing resin layer is provided on the first principal surface. The first sealing resin layer seals the first component. A marking portion is provided on a surface of the first sealing resin layer on a side opposite to the substrate. In the first sealing resin layer, the content rate of the filler is smaller in a second portion on the side opposite to the substrate than in a first portion on the substrate.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2002/0177294	12/2001	Otaki	N/A	N/A
2014/0284775	12/2013	Nomura	257/659	H01L 24/85
2018/0077802	12/2017	Kidoguchi	N/A	H05K 1/0298
2019/0074431	12/2018	Hasegawa	N/A	H01L 21/568
2020/0375022	12/2019	Nomura	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H01-120091	12/1988	JP	N/A
2000-022052	12/1999	JP	N/A
2001-044304	12/2000	JP	N/A
2002-353347	12/2001	JP	N/A
2015-015498	12/2014	JP	N/A
2018-019067	12/2017	JP	N/A
2018/116728	12/2017	WO	N/A
2019/159913	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report for International Patent Application No. PCT/JP2020/031103 dated Nov. 17, 2020. cited by applicant

Written Opinion for International Patent Application No. PCT/JP2020/031103 dated Nov. 17, 2020. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a continuation of International Application No. PCT/JP2020/031103 filed on Aug. 18, 2020 which claims priority from Japanese Patent Application No. 2019-171766 filed on Sep. 20, 2019 and Japanese Patent Application No. 2020-122590 filed on Jul. 17, 2020. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND OF THE DISCLOSURE
Field of the Disclosure
(1) The present disclosure relates to a module.
Description of the Related Art
(2) Japanese Patent Laying-Open No. 2015-15498 (PTL 1) discloses a configuration of a module. The module disclosed in PTL 1 includes a wiring substrate, a semiconductor element, a molding resin, and a shield layer. The semiconductor element is mounted on the wiring substrate. The mold resin seals the semiconductor element. The shield layer is provided on the mold resin. The molding resin has a marking on a surface by laser irradiation. The shield layer is provided on the molding resin having the marking. PTL 1: Japanese Patent Laying-Open No. 2015-15498
BRIEF SUMMARY OF THE DISCLOSURE
(3) In the conventional module, sometimes a filler made of an inorganic oxide is added to a sealing resin layer. However, when the filler is added to the sealing resin layer while a thickness of the sealing resin layer is reduced in order to reduce a height of the module, sometimes the marking laser applied to the sealing resin layer passes through the sealing resin layer through the filler. When the transmitted laser reaches a component mounted on the substrate sealed by the sealing resin layer, the laser may damage the component.
(4) The present disclosure has been made in view of the above problems, and an object of the present disclosure is to provide a module capable of reducing the height while preventing the damage to the component mounted on the substrate by the irradiation with the marking laser.
(5) A module according to the present disclosure includes a substrate, a first component, and a first sealing resin layer. The substrate includes a first principal surface. The first component is mounted on the first principal surface. The first sealing resin layer contains a filler containing an inorganic oxide as a main component. The first sealing resin layer is provided on the first principal surface. The first sealing resin layer seals the first component. A marking portion is provided on a surface of the first sealing resin layer on a side opposite to the substrate. In the first sealing resin layer, a content rate of the filler is smaller in a second portion on the side opposite to the substrate than in a first portion on the substrate.
(6) According to the present disclosure, the height of the module can be reduced while the damage to the component mounted on the substrate by the irradiation with the marking laser is prevented.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) FIG. **1** is a sectional view illustrating a module according to a first embodiment of the present disclosure.
(2) FIG. **2** is a flowchart illustrating a method for manufacturing the module of the first embodiment of the present disclosure.

(3) FIG. **3** is a sectional view illustrating a state in which an aggregate substrate is prepared in the method for manufacturing the module of the first embodiment of the present disclosure.
(4) FIG. **4** is a sectional view illustrating a state in which a first component is mounted on the aggregate substrate in the method for manufacturing the module of the first embodiment of the present disclosure.
(5) FIG. **5** is a sectional view illustrating a state in which a part of a first sealing resin layer is disposed on a first principal surface of the aggregate substrate in the method for manufacturing the module of the first embodiment of the present disclosure.
(6) FIG. **6** is a sectional view illustrating a state in which another part of the first sealing resin layer is disposed on a part of the first sealing resin layer on a side opposite to the aggregate substrate in the method for manufacturing the module of the first embodiment of the present disclosure.
(7) FIG. **7** is a sectional view illustrating a state in which a marking portion is provided by irradiating the first sealing resin layer with a laser beam in the method for manufacturing the module of the first embodiment of the present disclosure.
(8) FIG. **8** is a sectional view illustrating a state in which the aggregate substrate is divided to obtain a plurality of modules in the method for manufacturing the module of the first embodiment of the present disclosure.
(9) FIG. **9** is a sectional view illustrating a module according to a second embodiment of the present disclosure.
(10) FIG. **10** is a sectional view illustrating a module according to a third embodiment of the present disclosure.
(11) FIG. **11** is a partially enlarged view of an XI portion in FIG. **10**.
(12) FIG. **12** is a sectional view illustrating a module according to a fourth embodiment of the present disclosure.
(13) FIG. **13** is a sectional view illustrating a module according to a fifth embodiment of the present disclosure.
(14) FIG. **14** is a sectional view illustrating a module according to a sixth embodiment of the present disclosure.
(15) FIG. **15** is a sectional view illustrating a module according to a seventh embodiment of the present disclosure.
(16) FIG. **16** is a sectional view illustrating a module according to an eighth embodiment of the present disclosure.
(17) FIG. **17** is a sectional view illustrating a module according to a ninth embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE DISCLOSURE

(18) With reference to the drawings, a module according to each embodiment of the present disclosure will be described below. In the following description of each embodiment, the same or corresponding portion in the drawings is denoted the same reference numeral, and the description will not be repeated. In the present specification, the “color” of the material refers to a color exhibited in a visible light region.

First Embodiment

(19) FIG. **1** is a sectional view illustrating a module according to a first embodiment of the present disclosure. As illustrated in FIG. **1**, a module **100** of the first embodiment of the present disclosure includes a substrate **110**, a first component **120**, and a first sealing resin layer **130**.
(20) Substrate **110** includes a first principal surface **111** and a second principal surface **112**. Second principal surface **112** is located on the side opposite to first principal surface **111**. Substrate **110** has a peripheral side surface **114** extending from a peripheral end **113** of first principal surface **111** toward second principal surface **112**. Peripheral side surface **114** connects first principal surface **111** and second principal surface **112**.
(21) In the first embodiment, a plurality of external terminals **115** are provided on the second

principal surface. For example, external terminal **115** is a solder bump.
(22) First component **120** is mounted on first principal surface **111**. In the first embodiment, a plurality of first components **120** are mounted as first components **120**. Among the plurality of first components **120**, for example, a first component **121** is an integrated circuit (IC), and a first component **122** is an inductor or a capacitor.
(23) First sealing resin layer **130** is provided on first principal surface **111**. First sealing resin layer **130** seals first component **120**. Other components may be provided on first principal surface **111**. The other components may be sealed by first sealing resin layer **130** or exposed from first sealing resin layer **130**.
(24) First sealing resin layer **130** includes a filler containing an inorganic oxide as a main component. In the first embodiment, the filler is granular, specifically, spherical. For example, SiO.sub.2 or Al.sub.2O.sub.3 can be cited as the inorganic oxide. The “filler containing the inorganic oxide as a main component” refers to a state in which the inorganic oxide occupies half or more of the filler by weight. In the first embodiment, the filler is made of the inorganic oxide, and the inorganic oxide is SiO.sub.2, more specifically silica glass.
(25) First sealing resin layer **130** has improved adhesion and increased strength by including the filler. The filler is preferably included in first sealing resin layer **130** such that at least the first portion of first sealing resin layer **130** on the side of substrate **110** approaches a linear expansion coefficient substantially equal to that of first component **120**. The filler made of SiO.sub.2 in the first embodiment transmits the laser beam.
(26) In first sealing resin layer **130**, a maximum diameter of the filler is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**. In first sealing resin layer **130**, the maximum diameter of the filler included in the first portion on the side of substrate **110** is preferably greater than or equal to 20 μm and less than or equal to 30 μm. Thus, first sealing resin layer **130** can be configured to have a thermal expansion coefficient substantially equal to that of first component **120**. In first sealing resin layer **130**, the maximum diameter of the filler included in the second portion on the side opposite to substrate **110** is preferably less than or equal to 7 μm. As the filler is smaller, the transmission of the laser in first sealing resin layer **130** is prevented at the time of forming a later-described marking portion by irradiating first sealing resin layer **130** with the laser, and the influence of the laser on first component **120** is reduced. Accordingly, the thickness of first sealing resin layer **130** is not required to increase in order to prevent the laser transmission in first sealing resin layer **130**, and as a result, first sealing resin layer **130** can be thinned. In the first embodiment, the second portion occupies the entire portion of first sealing resin layer **130** on the side opposite to substrate **110**.
(27) In first sealing resin layer **130**, a content rate of the filler is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**. In first sealing resin layer **130**, the content rate of the filler contained in the first portion on the side of substrate **110** is preferably greater than or equal to 70 mass % and less than or equal to 90 mass %. Thus, first sealing resin layer **130** can be configured to have a thermal expansion coefficient substantially equal to that of first component **120**. In first sealing resin layer **130**, the content rate of the filler contained in the second portion on the side opposite to substrate **110** is preferably as small as possible, and most preferably zero. As the content rate of the filler decreases, the laser transmission in first sealing resin layer **130** is prevented at the time of forming the later-described marking portion by irradiating first sealing resin layer **130** with the laser, and the influence of the laser on first component **120** is reduced. Accordingly, the thickness of first sealing resin layer **130** is not required to increase in order to prevent the laser transmission in first sealing resin layer **130**, and as a result, first sealing resin layer **130** can be thinned.
(28) In the first embodiment, the content rate of the filler and the maximum diameter in first sealing resin layer **130** can be measured by observing a cut surface obtained by cutting first sealing resin layer **130** included in module **100** using a scanning electron microscope (SEM) or the like. For

example, the content rate can be calculated from the sum of the sectional areas of the fillers occupying the entire cut surface of first sealing resin layer **130**, the density of first sealing resin layer **130**, and the density of the fillers. The content rate of the filler in first sealing resin layer **130** may be calculated by collecting a test piece from first sealing resin layer **130** and measuring the weight of the test piece and the weight of the filler in the test piece obtained by burning the test piece to evaporate the resin component.

(29) In the first embodiment, first sealing resin layer **130** is laminated on substrate **110** and includes a plurality of resin layers having different content rates of fillers made of inorganic oxides from each other. In the first embodiment, the plurality of resin layers include a base layer **131** and a marking layer **132**. The plurality of resin layers may include a layer other than base layer **131** and the marking layer, but in the first embodiment, the plurality of resin layers are made of base layer **131** and marking layer **132**. First sealing resin layer **130** may be one layer containing a single resin material.

(30) In the first embodiment, base layer **131** is disposed on first principal surface **111** of substrate **110**, and marking layer **132** is disposed on base layer **131**. That is, in the first embodiment, the first portion of first sealing resin layer **130** on the side of substrate **110** corresponds to base layer **131**, and the second portion of first sealing resin layer **130** on the side opposite to substrate **110** corresponds to marking layer **132**.

(31) In the first embodiment, base layer **131** seals first component **120**, and first component **120** may be exposed from base layer **131**. When first component **120** is exposed from base layer **131**, the layer other than base layer **131** among the plurality of resin layers only has to seal first component **120**. For example, marking layer **132** may seal first component **120**.

(32) Marking layer **132** may be made of a filler-less resin containing no filler made of the inorganic oxide. The thickness of the thinnest portion of marking layer **132** is not particularly limited, but when marking layer **132** does not contain the filler, the thickness of the thinnest portion of marking layer **132** is preferably greater than or equal to 5 μm and less than or equal to 10 μm. When marking layer **132** includes the filler made of the inorganic oxide, the thickness dimension of the thinnest portion of marking layer **132** is preferably larger than the maximum diameter dimension of the filler made of the inorganic oxide.

(33) First sealing resin layer **130** may contain a conductive filler in addition to the filler made of the inorganic oxide. For example, carbon black can be cited as the conductive filler.

(34) In the first embodiment, among the plurality of resin layers constituting first sealing resin layer **130**, at least one layer located on the side opposite to substrate **110** of first component **120** is colored and thus has a light shading property. In the first embodiment, the at least one layer is black, and may have other colors. Among the plurality of resin layers, layers other than the at least one layer may also be colored. In the first embodiment, because both base layer **131** and marking layer **132** are colored, they have the light shading property. In the first embodiment, both base layer **131** and marking layer **132** are black.

(35) As described above, in the first embodiment, first sealing resin layer **130** is colored. However, first sealing resin layer **130** may be colored by the colored resin material constituting first sealing resin layer **130**, or colored by containing the conductive filler such as black carbon. When first sealing resin layer **130** is colored only by containing the conductive filler such as carbon black, the light shading property of first sealing resin layer **130** decreases as first sealing resin layer **130** becomes thinner. For this reason, in the first embodiment, the resin material contained in first sealing resin layer **130** is preferably colored. Thus, as compared with the case where first sealing resin layer **130** is colored only by containing the conductive filler, degradation of the light shading property can be prevented when first sealing resin layer **130** is thinned.

(36) A marking portion **133** is provided on a surface of first sealing resin layer **130** on the side opposite to substrate **110**. In the first embodiment, marking portion **133** is configured to be visually recognizable as a character, a figure, or a symbol when module **100** is viewed from the side of first

sealing resin layer **130**. Thus, module **100** can be distinguished from other modules. In the first embodiment, marking portion **133** is located so as to overlap at least one first component **120** when viewed from a laminating direction in which first sealing resin layer **130** is laminated on substrate **110**.

(37) Marking portion **133** has an outer shape of a recessed stripe. A depth of marking portion **133** is greater than or equal to 4 μm. When the depth of marking portion **133** is greater than or equal to 4 μm, marking portion **133** is easily visually recognized as the character, the figure, or the symbol.

(38) A method for manufacturing module **100** of the first embodiment of the present disclosure will be described below.

(39) FIG. **2** is a flowchart illustrating the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIG. **2**, the method for manufacturing the module of the first embodiment of the present disclosure includes a step S**1** of preparing the aggregate substrate, a step S**2** of mounting the first component, a step S**3** of disposing a part of the first sealing resin layer, a step S**4** of disposing another part of first sealing resin layer, a step S**5** of providing the marking portion, and a step S**6** of dividing the aggregate substrate to obtain the plurality of modules.

(40) FIG. **3** is a sectional view illustrating a state in which the aggregate substrate is prepared in the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIGS. **1** and **3**, in step S**1** of preparing an aggregate substrate **110***a*, aggregate substrate **110***a* is an aggregate of substrates **110** included in each of the plurality of modules **100**. Specifically, aggregate substrate **110***a* is in a state in which the plurality of substrates **110** are connected to each other on peripheral side surface **114**. Although external terminal **115** is previously formed on prepared aggregate substrate **110***a*, external terminal **115** may be formed in any of steps S**2** to S**5** when external terminal **115** is a solder bump.

(41) FIG. **4** is a sectional view illustrating a state in which the first component is mounted on the aggregate substrate in the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIG. **4**, in step S**2** of mounting first component **120**, the plurality of first components **120** are mounted on first principal surface **111** of aggregate substrate **110***a*.

(42) FIG. **5** is a sectional view illustrating a state in which a part of the first sealing resin layer is disposed on the first principal surface of the aggregate substrate in the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIG. **5**, in step S**3** of disposing a part of first sealing resin layer **130**, base layer **131** is disposed as a part of first sealing resin layer **130** on first principal surface **111** of aggregate substrate **110***a*. In the first embodiment, base layer **131** is disposed by being ground from the side opposite to first principal surface **111** after being laminated on first principal surface **111**. Thus, the thickness of base layer **131** can be reduced as much as possible while base layer **131** is configured to cover first component **120** at least on first principal surface **111**.

(43) FIG. **6** is a sectional view illustrating a state in which another part of the first sealing resin layer is disposed on a part of the first sealing resin layer on the side opposite to the aggregate substrate in the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIG. **6**, in step S**4** of disposing another part of first sealing resin layer **130**, marking layer **132** is disposed as another part of first sealing resin layer **130** on a base layer **131** that is a part of first sealing resin layer **130** on the side opposite to aggregate substrate **110***a*. In the first embodiment, marking layer **132** is disposed by applying the material constituting marking layer **132**.

(44) FIG. **7** is a sectional view illustrating a state in which the marking portion is provided by irradiating the first sealing resin layer with the laser beam in the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIG. **7**, in step S**5** of providing marking portion **133**, in the first embodiment, only marking layer **132** is irradiated with

the marking laser to form the plurality of marking portions **133**.

(45) In the irradiation with the marking laser, preferably a wavelength of the marking laser is appropriately changed in consideration of transmittance of first sealing resin layer **130**. In the first embodiment, for example, the wavelength of the marking laser is less than or equal to 532 nm. When the wavelength of the marking laser is less than or equal to 532 nm, the transmittance of the marking laser in first sealing resin layer **130** can be made relatively low. Thus, it is possible to prevent the marking laser from passing through first sealing resin layer **130** and reaching first component **120** to damage first component **120**. Consequently, the thickness of first sealing resin layer **130** can be reduced, whereby the height of module **100** can be reduced.

(46) FIG. **8** is a sectional view illustrating a state in which the aggregate substrate is divided to obtain a plurality of modules in the method for manufacturing the module of the first embodiment of the present disclosure. As illustrated in FIG. **8**, in step S**6** of dividing aggregate substrate **110***a* to obtain the plurality of modules **100**, aggregate substrate **110***a* is divided into pieces to obtain the plurality of modules **100**. Along with the division of aggregate substrate **110***a*, in the first embodiment, first sealing resin layer **130** is also divided.

(47) Through the above steps, module **100** of the first embodiment of the present disclosure as illustrated in FIG. **1** is manufactured.

(48) As described above, in module **100** of the first embodiment of the present disclosure, first sealing resin layer **130** contains the filler containing the inorganic oxide as the main component. First sealing resin layer **130** seals first component **120**. A marking portion **133** is provided on a surface of first sealing resin layer **130** on the side opposite to substrate **110**. In first sealing resin layer **130**, the content rate of the filler is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**.

(49) Thus, the height of module **100** can be reduced by reducing the thickness of first sealing resin layer **130** while the damage to first component **120** mounted on first principal surface **111** of substrate **110** by the irradiation with the marking laser is prevented.

(50) In the first embodiment, the filler made of the inorganic oxide is granular. In first sealing resin layer **130**, the maximum diameter of the filler made of the inorganic oxide is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**.

(51) Thus, the damage to first component **120** mounted on the first principal surface of substrate **110** by the irradiation with the marking laser can be suppressed, and the thickness of the second portion of first sealing resin layer **130** on the side opposite to substrate **110** can be further reduced.

(52) In the first embodiment, first sealing resin layer **130** is laminated on substrate **110** and includes a plurality of resin layers having different filler content rates from each other. Thus, a portion having a high content rate of the filler made of the inorganic oxide and a portion having a low content rate of the filler in first sealing resin layer **130** can be easily provided.

(53) In the first embodiment, among the plurality of resin layers, at least one layer located on the side opposite to substrate **110** of first component **120** has the light shading property. Thus, first component **120** can be prevented from being visually recognized through first sealing resin layer **130** and the visibility of the letter, the figure, or the symbol formed by marking portion **133** can be improved. At the same time, in first sealing resin layer **130**, the layer located on the side opposite to substrate **110** of first component **120** can be thinned.

Second Embodiment

(54) A module according to a second embodiment of the present disclosure will be described below. The module of the second embodiment of the present disclosure is mainly different from module **100** of the first embodiment of the present disclosure in that the component is mounted on second principal surface **112**. Accordingly, the description of the same configuration as that of module **100** of the first embodiment of the present disclosure will not be repeated.

(55) FIG. **9** is a sectional view illustrating the module of the second embodiment of the present disclosure. As illustrated in FIG. **9**, a module **200** of the second embodiment of the present

disclosure further includes a second component **240**, a second sealing resin layer **250**, and a connection electrode **260**. Second component **240** is mounted on second principal surface **112**. Second sealing resin layer **250** is provided on second principal surface **112** and seals second component **240**. Connection electrode **260** is connected to substrate **110** and located so as to penetrate second sealing resin layer **250**.

(56) Also in the second embodiment, because the content rate of the filler in first sealing resin layer **130** is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**, similarly to the first embodiment of the present disclosure, the height of module **100** can be reduced by reducing the thickness of first sealing resin layer **130** while the damage to first component **120** mounted on first principal surface **111** of substrate **110** by the irradiation with the marking laser is prevented.

Third Embodiment

(57) A module according to a third embodiment of the present disclosure will be described below. A module of the third embodiment of the present disclosure is different from module **100** of the first embodiment of the present disclosure in further including a shield layer. Accordingly, the description of the same configuration as that of module **100** of the first embodiment of the present disclosure will not be repeated.

(58) FIG. **10** is a sectional view illustrating the module of the third embodiment of the present disclosure. FIG. **11** is a partially enlarged view of an XI portion in FIG. **10**.

(59) As illustrated in FIG. **10**, a module **300** of the third embodiment of the present disclosure further includes a shield layer **370**. Shield layer **370** is located so as to cover the surface of first sealing resin layer **130**. Shield layer **370** is located along the outer shape of marking portion **133**. In addition, because shield layer **370** contains metal as described later, the surface orthogonal to the optical axis reflects light as it is in shield layer **370** and becomes bright, and the surface not orthogonal to the optical axis reflects light in another direction and becomes dark, so that brightness and darkness are easily realized according to an uneven shape of marking portion **133**. Thus, the visibility of the shape of marking portion **133** is improved by shield layer **370** located along the shape of marking portion **133**.

(60) As illustrated in FIG. **11**, shield layer **370** includes at least a conductive layer **371**. Thus, in the third embodiment, interference with first component **120** due to an electromagnetic wave from the outside can be prevented.

(61) In the third embodiment, in order to bring shield layer **370** and first sealing resin layer **130** into close contact with each other, the second portion of first sealing resin layer **130** on the side opposite to substrate **110** contains the filler made of the inorganic oxide. Specifically, marking layer **132** contains the filler made of the inorganic oxide.

(62) In the third embodiment, conductive layer **371** is preferably formed of a metal having high electrical conductivity. For example, conductive layer **371** is formed of Cu.

(63) As illustrated in FIG. **11**, in the third embodiment, shield layer **370** further includes a rust prevention layer **372** located in conductive layer **371** on the opposite side of first sealing resin layer **130**. Accordingly, in the third embodiment, oxidation or corrosion of conductive layer **371** can be prevented.

(64) In the third embodiment, for example, rust prevention layer **372** is formed of Ni, Cr, Ti, or an alloy of at least two metals selected from these. The alloy includes SUS. The average thickness of rust prevention layer **372** is thinner than that of conductive layer **371**. For example, the average thickness dimension of rust prevention layer **372** is larger than 0.1 μm and smaller than 10 μm.

(65) In the third embodiment, shield layer **370** further includes an adhesion layer **373** in contact with first sealing resin layer **130**. Thus, in the third embodiment, the adhesion between shield layer **370** and first sealing resin layer **130** can be improved.

(66) For example, adhesion layer **373** is formed of the same material as the material capable of forming rust prevention layer **372**. The average thickness of adhesion layer **373** is thinner than that

of conductive layer **371**.

(67) As illustrated in FIG. **10**, in the third embodiment, shield layer **370** is located so as to cover peripheral side surface **114**. Substrate **110** includes a ground electrode **316**. Ground electrode **316** is exposed on peripheral side surface **114** and electrically connected to shield layer **370**.

(68) The method for manufacturing module **300** of the third embodiment of the present disclosure further includes a step of providing shield layer **370** on first sealing resin layer **130** and peripheral side surface **114** of substrate **110** in addition to the step included in the method for manufacturing module **100** of the first embodiment of the present disclosure. Specifically, shield layer **370** is formed by a physical film forming method such as sputtering in a vacuum device. In order to improve the adhesion of shield layer **370** to first sealing resin layer **130** before the formation of shield layer **370**, the method for manufacturing module **300** of the third embodiment may further include a surface treatment step of first sealing resin layer **130**. Specifically, the surface treatment of first sealing resin layer **130** is performed by irradiating first sealing resin layer **130** with an ion such as Ar or N.

Fourth Embodiment

(69) A module according to a fourth embodiment of the present disclosure will be described below. The module of the fourth embodiment of the present disclosure is mainly different from module **300** of the third embodiment of the present disclosure in that the component is mounted on second principal surface **112**. Accordingly, the description of the same configuration as that of module **300** of the third embodiment of the present disclosure will not be repeated.

(70) FIG. **12** is a sectional view illustrating the module of the fourth embodiment of the present disclosure. A module **400** of the fourth embodiment of the present disclosure further includes second component **240**, second sealing resin layer **250**, and connection electrode **260** similarly to the module of the second embodiment of the present disclosure. In the fourth embodiment, shield layer **370** is also located on a side surface **451** of second sealing resin layer **250** from on peripheral side surface **114** of substrate **110**.

(71) Also in the fourth embodiment, because the content rate of the filler in first sealing resin layer **130** is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**, similarly to the first embodiment of the present disclosure, the height of module **100** can be reduced by reducing the thickness of first sealing resin layer **130** while the damage to first component **120** mounted on first principal surface **111** of substrate **110** by the irradiation with the marking laser is prevented.

Fifth Embodiment

(72) A module according to a fifth embodiment of the present disclosure will be described below. The module of the fifth embodiment of the present disclosure is mainly different from module **300** of the third embodiment of the present disclosure in that the first sealing resin is partially formed on the first principal surface of the substrate, a connector is provided in a portion of the first principal surface where the first sealing resin layer is not formed, and an antenna is provided on the second principal surface of the substrate. Accordingly, the description of the same configuration as that of module **300** of the third embodiment of the present disclosure will not be repeated.

(73) FIG. **13** is a sectional view illustrating the module of the fifth embodiment of the present disclosure. As illustrated in FIG. **13**, in a module **500** of the fifth embodiment of the present disclosure, first sealing resin layer **130** is partially formed on a first principal surface **511** of the substrate. Shield layer **370** is located so as to cover the surface of first sealing resin layer **130** and peripheral side surface **114**, but shield layer **370** does not cover at least a part of the region where first sealing resin layer **130** is not formed on first principal surface **511**. Specifically, shield layer **370** does not cover the region where first sealing resin layer **130** is not formed on first principal surface **511**. In first principal surface **511**, connector **523** connected to substrate **110** is provided in the region where first sealing resin layer **130** is not formed. Furthermore, an antenna **541** connected to substrate **110** is provided on second principal surface **112** of substrate **110**.

(74) Also in the fifth embodiment, because the content rate of the filler in first sealing resin layer **130** is smaller in the second portion on the side opposite to substrate **110** than in the first portion on the side of substrate **110**, similarly to the first embodiment of the present disclosure, the height of module **500** can be reduced by reducing the thickness of first sealing resin layer **130** while the damage to first component **120** mounted on first principal surface **511** of substrate **110** by the irradiation with the marking laser is prevented.

Sixth Embodiment

(75) A module according to a sixth embodiment of the present disclosure will be described below. The module of the sixth embodiment of the present disclosure is mainly different from module **100** of the first embodiment of the present disclosure in the configuration of the first sealing resin layer. Accordingly, the description of the same configuration as that of module **100** of the first embodiment of the present disclosure will not be repeated.

(76) FIG. **14** is a sectional view illustrating the module of the sixth embodiment of the present disclosure. As illustrated in FIG. **14**, in a module **600** of the sixth embodiment of the present disclosure, the second portion is provided in a part of a first sealing resin layer **630** as viewed from the laminating direction in which first sealing resin layer **630** is laminated on substrate **110**. Thus, in the portion of first sealing resin layer **630** where the second portion is not provided, the height of module **600** can be further reduced.

(77) Specifically, in first sealing resin layer **630**, marking layer **132** is provided on a part of base layer **131** on the side opposite to substrate **110** when viewed from the base layer **131**.

(78) Module **600** of the sixth embodiment further includes a third component **620** mounted on first principal surface **111**. Third component **620** is exposed from the first portion of first sealing resin layer **630** on the side opposite to substrate **110**. Thus, the height of module **600** can be further reduced in the portion of module **600** where third component **620** is provided.

(79) Specifically, third component **620** is exposed from base layer **131** on the side opposite to substrate **110**. That is, third component **620** is located so as not to overlap the second portion (marking layer **132**) when viewed from the laminating direction. Third component **620** may be sealed in first sealing resin layer **630**. Third component **620** is a component similar to the component that can be used as first component **120**. In the sixth embodiment, specifically third component **620** is an IC. Third component **620** may be a filter component such as a surface acoustic wave filter.

Seventh Embodiment

(80) A module according to a seventh embodiment of the present disclosure will be described below. The module of the seventh embodiment of the present disclosure is mainly different from module **200** of the second embodiment of the present disclosure in the configuration of the first sealing resin layer. Accordingly, the description of the same configuration as that of module **200** of the second embodiment of the present disclosure will not be repeated.

(81) FIG. **15** is a sectional view illustrating the module of the seventh embodiment of the present disclosure. As illustrated in FIG. **15**, first sealing resin layer **630** is provided in a module **700** of the seventh embodiment of the present disclosure similarly to module **600** of the sixth embodiment of the present disclosure. That is, the second portion (marking layer **132**) is provided in a part of first sealing resin layer **630** when viewed from the laminating direction in which first sealing resin layer **630** is laminated on substrate **110**. Thus, the height of module **700** can be further reduced in the portion of first sealing resin layer **630** where the second portion is not provided.

(82) Module **700** of the seventh embodiment of the present disclosure further includes third component **620** similar to the module of the sixth embodiment of the present disclosure. Third component **620** is exposed from the first portion (base layer **131**) of first sealing resin layer **630** on the side opposite to substrate **110**. Thus, the height of module **700** can be further reduced in the portion of module **700** where third component **620** is provided.

Eighth Embodiment

(83) A module according to an eighth embodiment of the present disclosure will be described below. The module of the eighth embodiment of the present disclosure is mainly different from module **300** of the third embodiment of the present disclosure in the configuration of the first sealing resin layer. Accordingly, the description of the same configuration as that of the module of the third embodiment of the present disclosure will not be repeated.
(84) FIG. **16** is a sectional view illustrating the module of the eighth embodiment of the present disclosure. As illustrated in FIG. **16**, first sealing resin layer **630** is provided in a module **800** of the eighth embodiment of the present disclosure similarly to module **600** of the sixth embodiment of the present disclosure. That is, the second portion (marking layer **132**) is provided in a part of first sealing resin layer **630** when viewed from the laminating direction in which first sealing resin layer **630** is laminated on substrate **110**. Thus, the height of module **800** can be further reduced in the portion of first sealing resin layer **630** where the second portion is not provided.
(85) Module **800** of the eighth embodiment of the present disclosure further includes third component **620** similar to the module of the sixth embodiment of the present disclosure. Third component **620** is exposed from the first portion (base layer **131**) of first sealing resin layer **630** on the side opposite to substrate **110**. Thus, the height of module **800** can be further reduced in the portion of module **800** where third component **620** is provided.
(86) Furthermore, in the eighth embodiment, shield layer **370** is in contact with the surface of third component **620** at the portion where third component **620** is exposed from first sealing resin layer **630**. For example, shield layer **370** contains Cu, and thus has higher thermal conductivity than that of first sealing resin layer **630**. Accordingly, in third component **620**, heat dissipation to the outside of module **800** is further improved.
Ninth Embodiment
(87) A module according to a ninth embodiment of the present disclosure will be described below. The module of the ninth embodiment of the present disclosure is mainly different from module **400** of the fourth embodiment of the present disclosure in the configuration of the first sealing resin layer. Accordingly, the description of the same configuration as that of the module of the fourth embodiment of the present disclosure will not be repeated.
(88) FIG. **17** is a sectional view illustrating the module of the ninth embodiment of the present disclosure. As illustrated in FIG. **17**, first sealing resin layer **630** is provided in a module **900** of the ninth embodiment of the present disclosure similarly to module **600** of the sixth embodiment of the present disclosure. That is, the second portion (marking layer **132**) is provided in a part of first sealing resin layer **630** when viewed from the laminating direction in which first sealing resin layer **630** is laminated on substrate **110**. Thus, the height of module **900** can be further reduced in the portion of first sealing resin layer **630** where the second portion is not provided.
(89) Module **900** of the ninth embodiment of the present disclosure further includes third component **620** similar to the module of the sixth embodiment of the present disclosure. Third component **620** is exposed from the first portion (base layer **131**) of first sealing resin layer **630** on the side opposite to substrate **110**. Thus, the height of module **900** can be further reduced in the portion of module **900** where third component **620** is provided.
(90) Furthermore, in the ninth embodiment, shield layer **370** is in contact with the surface of third component **620** at the portion where third component **620** is exposed from first sealing resin layer **630**. For example, shield layer **370** contains Cu, and thus has higher thermal conductivity than that of first sealing resin layer **630**. Accordingly, in third component **620**, heat dissipation to the outside of module **900** is further improved.
(91) In the description of the above embodiments, configurations that can be combined may be combined with each other.
(92) It should be considered that the disclosed embodiment is illustrative and non-restrictive in every respect. The scope of the present disclosure is defined by not the description above, but the claims, and it is intended that all modifications within the meaning and scope equivalent to the

claims are included in the present disclosure. **100**, **200**, **300**, **400**, **500**, **600**, **700**, **800**, **900**: module, **110**: substrate, **110***a*: aggregate substrate, **111**, **511**: first principal surface, **112**: second principal surface, **113**: peripheral end, **114**: peripheral side surface, **115**: external terminal, **120**, **121**, **122**: first component, **130**, **630**: first sealing resin layer, **131**: base layer, **132**: marking layer, **133**: marking portion, **240**: second component, **250**: second sealing resin layer, **260**: connection electrode, **316**: ground electrode, **370**: shield layer, **371**: conductive layer, **372**: rust prevention layer, **373**: adhesion layer, **451**: side surface, **523**: connector, **541**: antenna, and **620**: third component.

Claims

1. A module comprising: a substrate including a first principal surface; a first component mounted on the first principal surface; and a first sealing resin layer containing a filler, provided on the first principal surface, and sealing the first component, wherein the filler contains an inorganic oxide as a main component, wherein the first sealing resin layer includes a first portion disposed on the substrate, and a second portion disposed on a side of the first portion opposite to the substrate, a marking portion is provided on a surface of the second portion, in the first sealing resin layer, a content rate of the filler is lower in the second portion than in the first portion, and the first sealing resin layer is laminated on the substrate and includes a plurality of resin layers having different content rates of the filler from each other.
2. The module according to claim 1, wherein the filler is granular, and in the first sealing resin layer, a maximum diameter of the filler is smaller in the second portion than in the first portion.
3. The module according to claim 2, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, and wherein the module further comprises: a second component mounted on the second principal surface; a second sealing resin layer provided on the second principal surface to seal the second component; and a connection electrode connected to the substrate and located so as to penetrate the second sealing resin layer.
4. The module according to claim 2, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the shield layer includes at least a conductive layer.
5. The module according to claim 2, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, the first sealing resin layer is partially provided on the first principal surface of the substrate, a connector connected to the substrate is provided in a region of the first principal surface where the first sealing resin layer is not provided, and an antenna connected to the substrate is provided on the second principal surface of the substrate.
6. The module according to claim 1, wherein at least one layer of the plurality of resin layers located on a side of the first component opposite to a substrate side of the first component has a light shading property.
7. The module according to claim 6, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, and wherein the module further comprises: a second component mounted on the second principal surface; a second sealing resin layer provided on the second principal surface to seal the second component; and a connection electrode connected to the substrate and located so as to penetrate the second sealing resin layer.
8. The module according to claim 6, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the shield layer includes at least a conductive layer.
9. The module according to claim 1, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, and wherein the module further comprises: a second component mounted on the second principal surface; a second sealing resin layer provided on the second principal surface to seal the second component; and a connection

electrode connected to the substrate and located so as to penetrate the second sealing resin layer.
10. The module according to claim 9, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the shield layer is also located on a side surface of the second sealing resin layer from on a peripheral side surface of the substrate.
11. The module according to claim 9, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the shield layer includes at least a conductive layer.
12. The module according to claim 1, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the shield layer includes at least a conductive layer.
13. The module according to claim 1, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, the first sealing resin layer is partially provided on the first principal surface of the substrate, a connector connected to the substrate is provided in a region of the first principal surface where the first sealing resin layer is not provided, and an antenna connected to the substrate is provided on the second principal surface of the substrate.
14. The module according to claim 1, wherein the second portion is provided in a part of the first sealing resin layer when viewed from a laminating direction in which the first sealing resin layer is laminated on the substrate.
15. The module according to claim 14, further comprising a third component mounted on the first principal surface, wherein the third component is exposed from the first portion of the first sealing resin layer on the side of the first portion opposite to the substrate.
16. The module according to claim 1, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, and wherein the module further comprises an external terminal provided on the second principal surface.
17. The module according to claim 1, wherein the substrate further includes a second principal surface located on a side opposite to the first principal surface, and wherein the module further comprises: a second component mounted on the second principal surface; a second sealing resin layer provided on the second principal surface to seal the second component; and a connection electrode connected to the substrate and located so as to penetrate the second sealing resin layer.
18. The module according to claim 1, further comprising a shield layer located so as to cover a surface of the first sealing resin layer, wherein the shield layer includes at least a conductive layer.